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ORIGINAL ARTICLE

Maximal and explosive strength training elicit distinct neuromuscular adaptations, specific to the training stimulus

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Abstract
Purpose To compare the effects of short-term maximal (MST) vs. explosive (EST) strength training on maximal and explosive force production, and assess the neural adaptations underpinning any training-specific functional changes.
Methods Male participants completed either MST ($n = 9$) or EST ($n = 10$) for 4 weeks. In training participants were instructed to: contract as fast and hard as possible for ~1 s (EST); or contract progressively up to 75 % maximal voluntary force (MVF) and hold for 3 s (MST). Pre- and post-training measurements included recording MVF during maximal voluntary contractions and explosive force at 50-ms intervals from force onset during explosive contractions. Neuromuscular activation was assessed by recording EMG RMS amplitude, normalised to a maximal M-wave and averaged across the three superficial heads of the quadriceps, at MVF and between 0–50, 0–100 and 0–150 ms during the explosive contractions.
Results Improvements in MVF were significantly greater ($P < 0.001$) following MST (+21 ± 12 %) than EST (+11 ± 7 %), which appeared due to a twofold greater increase in EMG at MVF following MST. In contrast, early phase explosive force (at 100 ms) increased following EST (+16 ± 14 %), but not MST, resulting in a time × group interaction effect ($P = 0.03$), which appeared due to a greater increase in EMG during the early phase (first 50 ms) of explosive contractions following EST ($P = 0.052$).
Conclusions These results provide evidence for distinct neuromuscular adaptations after MST vs. EST that are specific to the training stimulus, and demonstrate the independent adaptability of maximal and explosive strength.

Keywords Resistance training · Rate of force development · Neural activation · Specificity

Abbreviations

ANOVA	Analysis of variance
$EMG_{0–50}$	Electromyography recorded during explosive contractions over a time period denoted in subscript
EMG_{MVF}	Electromyography recorded at MVF
EST	Explosive strength training
F_{50}	Explosive force recorded at a discrete time point from force onset denoted in subscript
M_{max}	Maximal M-wave
MST	Maximal strength training
MVC	Maximal voluntary contraction
MVF	Maximal voluntary force
M-wave	Compound muscle action potential
RF	Rectus femoris
RMS	Root mean square
VL	Vastus lateralis
VM	Vastus medialis

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Introduction

Skeletal muscles apply force to the skeleton to move and/or stabilise the body, and thus their capacity for volitional force production (skeletal muscle strength) limits function. Two components of skeletal muscle strength include: maximal strength, measured as the maximal voluntary force (MVF) the muscles can produce; and explosive strength, the ability of the muscles to increase force rapidly from a low force or resting state (Tillin et al. 2012b). Maximal strength sets the upper functional limit of the musculoskeletal system and is important for relatively slow movement tasks [e.g. ambulation/locomotion of older individuals (Faulkner et al. 2007) and neuromuscular patients (Ada et al. 2000), or sports activities such as rugby scrummaging (Quarrie and Wilson. 2000) and wrestling (Garcia-Pallares et al. 2011)]. However, it takes time to develop MVF [>100 ms in concentric contractions (Tillin et al. 2012a) and >250 ms in isometric and eccentric contractions (Thorstensson et al. 1976; Tillin et al. 2012a)], and thus explosive strength is considered more important where time available to develop force is limited [e.g. restabilising the body following a loss of balance (Domire et al. 2011; Pijnappels et al. 2008) and sports activities such as sprinting and jumping (de Ruiter et al. 2006; Tillin et al. 2013a)]. Understanding how these components of strength are affected by different strength training modalities has important practical implications for improving health and sports performance.

Time under tension and magnitude of the training load are considered important stimuli for developing maximal strength (Crewther et al. 2005), and thus maximal strength training typically involves sustained (>2 s) contractions against loads ≥70 % MVF (Del Balso and Cafarelli 2007; Jones and Rutherford 1987; Kubo et al. 2001). In contrast, training for explosive strength is typically characterised by a series of short (≤1 s) contractions performed as rapidly as possible (Barry et al. 2005; de Ruiter et al. 2012) to practise and improve the rate of force development. Based on the training principle of specificity, it is conceivable that these different training stimuli induce distinct functional adaptations, e.g. with greater gains in the explosive strength after explosive training. On the other hand, maximal and explosive strength are determined by similar physiological mechanisms [e.g. muscle size and neural drive; (Andersen and Aagaard 2006; Duchateau et al. 2006; Hakkinen and Keskinen 1989; Schantz et al. 1983; Tillin et al. 2010)], and might be expected to exhibit similar changes in response to training.

Evidence in support of distinct training stimuli for improvements in maximal and explosive strength is equivocal. Individual studies have examined the functional responses to either maximal strength training, explosive strength training, or a combination of both, with some investigations observing distinct effects on maximal and explosive strength (Andersen et al. 2010; Gruber et al. 2007; Rich and Cafarelli 2000; Tillin et al. 2011) and others reporting improvements in both attributes (Barry et al. 2005; de Ruiter et al. 2012; Del Balso and Cafarelli 2007; Kubo et al. 2001; Suetta et al. 2004; Tillin et al. 2012b). Methodological differences (e.g. training duration, intensity and volume; instructions to the participants during the training; and the type of training contractions performed) preclude meaningful comparisons of functional responses between separate studies. Therefore a direct comparison of maximal and explosive strength training is required to establish whether there are distinct functional responses to these different training stimuli.

Whilst maximal and explosive strength training might be expected to elicit distinct functional responses, the physiological bases of this specificity are unknown. The changes in function with any type of strength training are underpinned by neural and/or morphological adaptations (Folland and Williams 2007), with neural adaptations widely thought to dominate the adaptive response to short-term training (within 4 weeks; Folland and Williams 2007; Sale 2003). Increased maximal strength has been associated with increased neuromuscular activation at MVF (Del Balso and Cafarelli 2007; Tillin et al. 2011), whilst increased explosive strength has been accompanied by greater activation during the explosive phase of contraction (Barry et al. 2005; de Ruiter et al. 2012; Tillin et al. 2012b). However, the degree of specificity and/or transferability of these neural changes between the explosive and maximal phases of contraction, and vice versa, are unclear. Neuromuscular activation can be assessed by EMG amplitude normalised to a maximum evoked compound muscle action potential (M-wave), as a non-volitional reference, to reduce the between-subject variability and increase the sensitivity of the experiment to detect changes in activation (Buckthorpe et al. 2012). In addition, the proportional changes in neuromuscular activation during the explosive and maximum phases of contraction can be examined by assessing EMG during the explosive phase of contraction relative to EMG at MVF.

The purpose of this study was to compare the effects of short-term maximal vs. explosive strength training on maximal and explosive force production, and assess the neural adaptations underpinning any training-specific functional changes. We hypothesised that the two types of training would elicit distinct functional and neural changes specific to the training stimulus.

Methods

Participants

Nineteen male participants who were recreationally active (moderate exercise ≤4 times per week), but had not been involved in any form of lower-body strength training for

Table 1 Physical characteristics of the maximal and explosive strength training groups prior to the training

	Strength training group	
	Maximal	Explosive
Age (years)	20.9 ± 1.1	20.2 ± 2.4
Height (m)	1.82 ± 0.05	1.82 ± 0.08
Body mass (kg)	81.1 ± 6.8	73.6 ± 7.4**
MVF (N)	585 ± 84	482 ± 58**
MVF/body mass (N.kg^{-1})	7.2 ± 0.8	6.6 ± 0.8

MVF maximal voluntary force

** Denotes a significant difference between the groups ($P < 0.01$)

>6 months prior to the start of the study, were recruited and completed either maximal strength training (MST group; $n = 9$) or explosive strength training (EST group; $n = 10$). All participants were healthy, injury free and provided written informed consent prior to their involvement in this study, which was approved by the Loughborough University Ethical Advisory Committee. At baseline the groups were of similar age and height, and whilst they differed in body mass and MVF, there were no differences in MVF relative to body mass (Table 1).

Overview

Participants completed a familiarisation session and a measurement session (separated by 2–3 days) before, and one measurement session after, 4 weeks of unilateral isometric strength training of the knee extensors. The pre-training measurement session took place within 10 days prior to the start of training, whilst the post-training measurement session took place 2–3 days after the last training session. During the familiarisation session, participants practised the range of measurement tasks without data recording. Pre- and post-training measurement sessions involved recording external knee extension force and surface EMG of the superficial knee extensors during a series of explosive voluntary and maximal voluntary (MVCs) isometric contractions of the knee extensors. EMG responses (M-waves) to electrical stimulation of the femoral nerve with single, supramaximal impulses were also recorded for normalisation of the EMG signal recorded in the voluntary efforts. Training was of one leg chosen at random and involved isometric contractions of the knee extensors (either EST or MST) performed four times a week for 4 weeks with the same apparatus as the measurement sessions. Some within-group comparisons of the trained vs. untrained leg after maximal (Tillin et al. 2011) and explosive (Tillin et al. 2012b) strength training have previously been reported separately. Participants were instructed to maintain their normal physical activities throughout the period of the study.

Whilst dynamic contractions might be considered more relevant to functional human movement, we chose an isometric contraction model to assess the effects of MST and EST on these two components of strength as it provides a more experimentally controlled situation in which to assess the mechanisms underpinning the changes in function.

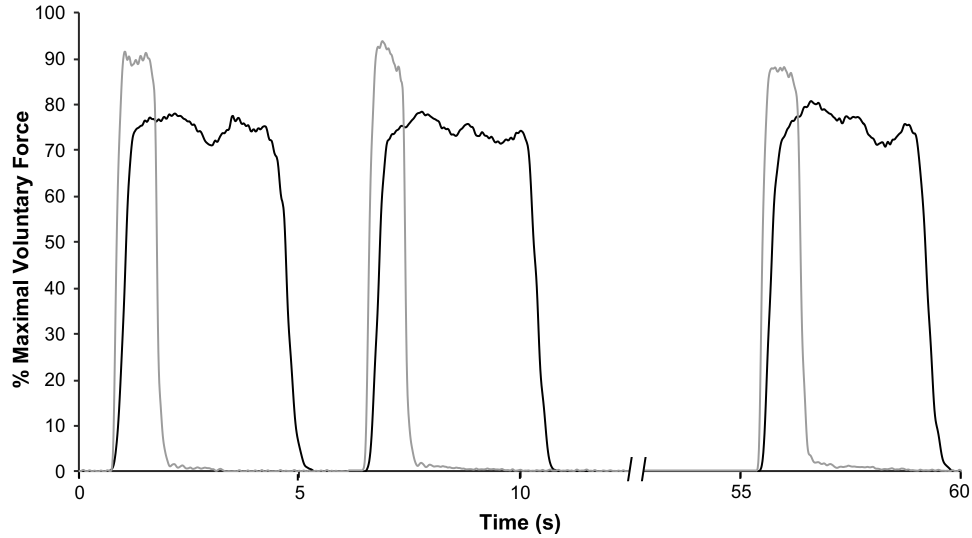
Training

Each training session consisted of a brief warm-up of submaximal isometric knee extensions, followed by four sets (separated by 2 min) of ten isometric contractions of the knee extensors (each set lasting ~60 s). However, the type of contractions performed differed between the training groups. The MST group were instructed to increase force over a 1-s period, up to 75 % of their MVF, hold for 3 s, and then relax for 2 s before completing the next contraction in the set. There was no specific instruction to produce force rapidly during the MST. In contrast, the EST group were instructed to contract as “fast and hard” as possible for ~1 s without producing a prior knee-flexor force, in an attempt to achieve at least 90 % of their MVF, and then relax for 5 s before completing the next contraction in the set. Thus, whilst the duration of loading was greater for the MST group, the peak loads and loading rates were greater for the EST group (Fig. 1). For biofeedback in the MST group, a computer monitor displayed the force–time curve with a horizontal cursor on 75 % MVF, whilst in the EST group the monitor displayed the slope of the force–time curve (established with a 1-ms constant epoch) and the baseline of the force–time curve that was used to confirm that no prior knee-flexor force had occurred. MVF was initially established in familiarisation and the pre-training measurement sessions (see below) and re-established at the start of the first training session each week.

Force and EMG recording

Measurement and training sessions were completed in an isometric strength testing chair (Bojsen-Moller et al. 2005), with knee and hip angles of 85° and 100°, respectively (180° representing full extension). Participants were secured firmly in the chair with a waist belt and shoulder straps. An ankle strap was consistently placed 3 cm proximal to the medial malleolus and was in series with a calibrated linear response strain gauge (Jones and Parker 1989) positioned perpendicular to the tibia. This low noise strain gauge [range of baseline noise <0.2 N or <0.02 % full scale deflection (Tillin et al. 2010)] consisted of a U-shaped aluminium beam with two high-frequency response silicon strain gauges bonded either side of the horizontal section. The force signal from the strain gauge was amplified (×500), sampled at 2,000 Hz using

Fig. 1 Example force–time curves recorded during the first (two) and last isometric knee extension contractions of one training set performed by participants in the maximal strength training (*black line*) and explosive strength training (*grey line*) groups. The duration of loading in each contraction was greater in the maximal strength training group (~4 vs. ~1 s), but the explosive strength training group experienced greater peak loads (~90 vs. 75 % maximal voluntary force) and rate of loading (slope of the force–time curve)



an external A/D converter (16-bit signal recording resolution; Micro 1401, CED, Cambridge, UK), interfaced with a personal computer using Spike 2 software (CED, Cambridge, UK), and digitally notch filtered at 100 and 200 Hz [q factor of 100; (Tillin et al. 2010)]. Baseline resting force was subtracted from all active force recordings to provide gravity correction.

Single differential EMG surface electrodes (Delsys Bagnoli-4, Boston, USA) were placed: over the belly of the rectus femoris (RF), vastus lateralis (VL), and vastus medialis (VM); parallel to the presumed orientation of the muscle fibres; and at ~50 % (RF), 55 % (VL), and 90 % (VM) of the distance between the greater trochanter and lateral femoral condyle. EMG signals were amplified (×100 differential amplifier 20–450 Hz), sampled at 2,000 Hz via the same A/D convertor and PC software as the force signal, and band-pass filtered (6–500 Hz) using a fourth-order, zero-lag Butterworth digital filter.

Pre- and post-training measurement sessions

M-wave recordings

M-waves were elicited by electrically stimulating the femoral nerve (DS7AH, Digitimer Ltd., UK) with square wave pulses (0.1 ms duration). The anode (carbon rubber, 7 × 10 cm; EMS Physio Ltd, Greenham, UK) was taped to the skin over the greater trochanter. The cathode, a custom-adapted stimulation probe (1-cm diameter, Electro Medical Supplies, Wantage, UK) protruding 2 cm perpendicular from the centre of a plastic base (4 × 5 cm), was taped to the skin over the femoral nerve in the femoral triangle. The precise location of the cathode was determined as the position that evoked the greatest twitch response for a particular submaximal electrical current (typically 30–50 mA).

The maximal knee extensor twitch response for a single electrical impulse was first established before eliciting three supramaximal (120 % of the current required to evoke a maximal twitch) twitch contractions at 12-s intervals. Peak–peak M-wave (M_{max}) responses of the RF, VL, and VM were averaged across the three supramaximal twitch contractions.

Explosive voluntary contractions

Following the evoked contractions and a warm-up of submaximal voluntary contractions, participants completed ten explosive voluntary contractions (each ~20 s apart). Prior to each contraction, participants were provided with the same instruction as that given during the training contractions performed by the EST group. Explosive voluntary contractions were completed separately from MVCs (detailed below) because previous work has reported the performance outcome of voluntary contractions to be specific to the instruction given to the participants (Sahaly et al. 2001). During off-line analysis, the three explosive voluntary contractions with the largest peak rate of force development (determined from a 1-ms moving epoch) and no change in baseline force >0.5 N during the 100 ms prior to contraction onset were used for analysis. Voluntary explosive force was measured at 50, 100, and 150 ms from force onset (F_{50}, F_{100}, F_{150}) and normalised to body mass to account for the group differences in body mass at the start of the study. Voluntary explosive force values are also expressed relative to MVF to assess how explosive and maximal strengths changed in proportion to each other following the two types of training.

To assess agonist neural drive, the root mean square (RMS) of the EMG signal of each muscle was measured over three time periods (0–50 ms, EMG_{0-50}; 0–100 ms,

Table 2 Maximal voluntary force (MVF) and explosive force (*F*) production (recorded at 50-ms intervals from force onset) pre- and post-training in the maximal (MST) and explosive strength training (EST) groups

	MST group		EST group		Interaction effect (*P* value)
	Pre	Post	Pre	Post	
MVF ($N.kg^{-1}$)	7.22 ± 0.80	8.70 ± 1.19**	6.58 ± 0.83	7.28 ± 0.71***	0.020
F_{50} ($N.kg^{-1}$)	1.64 ± 0.71	1.70 ± 0.72	1.23 ± 0.43	1.89 ± 0.65*	0.083
F_{100}($N.kg^{-1}$)	4.21 ± 0.78	4.34 ± 0.72	3.94 ± 0.61	4.53 ± 0.62**	0.030
F_{150} ($N.kg^{-1}$)	5.55 ± 0.76	5.96 ± 0.80	5.05 ± 0.70	5.71 ± 0.63***	0.235
F_{50} (%MVF)	23.1 ± 11.2	19.9 ± 9.3	19.0 ± 7.3	25.8 ± 8.1*	0.032
F_{100} (%MVF)	58.6 ± 10.8	50.4 ± 9.4**	59.9 ± 5.6	62.1 ± 5.2	0.002
F_{150} (%MVF)	77.2 ± 10.5	69.2 ± 10.7*	76.7 ± 4.5	78.5 ± 4.3	0.005

All variables were normalised to body mass, and explosive force variables are also expressed relative to MVF. Time (pre vs. post) × group (MST vs. EST) interaction effects are also reported. Data are mean ± SD

Within-group effects of training are denoted by * ($P < 0.05$), ** ($P < 0.01$), or *** ($P < 0.001$)

$EMG_{0–100}$; and 0–150 ms, $EMG_{0–150}$) from EMG onset (first agonist muscle to be activated). Agonist (RF, VL, and VM) RMS EMG values were normalised to M_{max} and averaged across the three muscles to give a mean agonist value. Agonist RMS EMG values recorded during the explosive contractions were also expressed relative to EMG at MVF (EMG_{MVF}; see below) and averaged across the three muscles.

All explosive voluntary force and EMG variables were averaged across the three explosive voluntary contractions chosen for analysis. Force and EMG onsets were identified manually by the same trained investigator using a systematic previously published protocol (Tillin et al. 2010, 2013b). Briefly, force and EMG signals were viewed on a constant *x* axis scale of 500 ms, and constant *y* axes scales of 1 N (force) and 10 mV (EMG). This provided a good resolution from which to establish the baseline noise pattern and interpolate onset, defined as the last peak/trough before the signal deflected away from the baseline. Manual identification is considered the 'gold standard' method for detecting signal onsets (Allison 2003; Moretti et al. 2003; Pain and Hibbs 2007; Pulkovski et al. 2008; Tillin et al. 2013b).

Maximal voluntary contractions

Following the explosive contractions, participants completed at least three knee extensor isometric MVCs (separated by ≥30 s), in which participants were instructed to push as hard as possible for 3–5 s. Biofeedback and verbal encouragement were provided during and between each MVC. Knee extensor MVF was the greatest instantaneous voluntary force achieved by a participant in any of the knee extensor MVCs or explosive contractions during that laboratory visit, and was normalised to body mass. RMS EMG of each agonist muscle at MVF (EMG_{MVF}) was recorded during a 500-ms epoch (250 ms either side of MVF), normalised to M_{max} (Sale 2003) and averaged across the three quadriceps muscles.

Statistical analysis

The influence of time (pre- vs. post-training) and group (MST vs. EST) on each dependent variable was assessed via a two-way repeated measures ANOVA with training and group as within- and between-participant factors, respectively. When an interaction or main-training effect occurred, paired *t* tests were used to assess the influence of training within each group. Statistical significance was set at $P < 0.05$. Statistical analysis was completed using SPSS version 19. All data are presented as mean ± standard deviation; apart from those in the figures where data are mean ± standard error of the mean for presentation purposes. Percentage change in each dependent variable was calculated as the mean percentage change of all participants within a group.

Results

Force variables

There was a main effect of time on MVF (ANOVA, $P < 0.001$), due to an increase in MVF (Table 2) after both MST (+21 ± 12 %) and EST (+11 ± 7 %). The increase in MVF was greater in the MST group resulting in a time × group interaction effect (ANOVA, $P = 0.02$; Table 2; Fig. 2).

There was a main effect of time on F_{50} (ANOVA, $P = 0.041$), F_{100} (ANOVA, $P = 0.002$), and F_{150} (ANOVA, $P < 0.001$). Post hoc analysis revealed no change in F_{50} or F_{100} following MST (Table 2), whilst EST increased

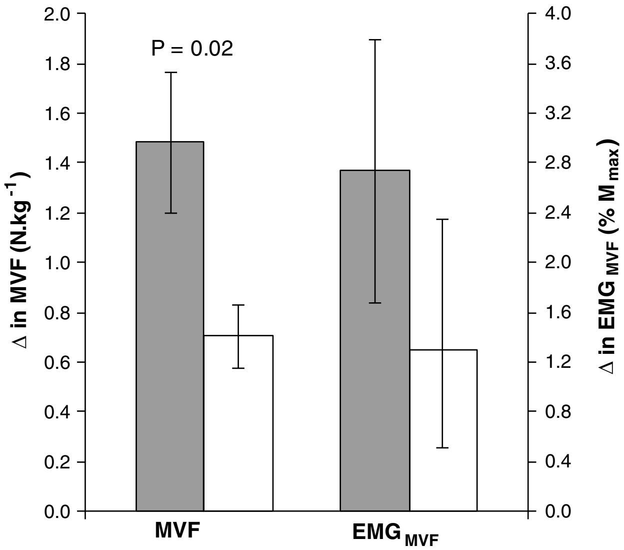


Fig. 2 Pre- to post-training changes in maximal voluntary force (MVF) and mean quadriceps EMG at MVF (EMG_{MVF}) during isometric knee extensions in the maximal (*filled bars*) and explosive (*open bars*) strength training groups. MVF and EMG_{MVF} were normalised to body mass and maximal M-wave (M_{max}), respectively, prior to calculating changes. The *P* value denotes a time × group interaction effect ($P < 0.05$) determined via a two-way ANOVA (time, pre vs. post; by group, maximal vs. explosive). Data are mean ± SEM

both of these variables (F_{50}, +70 ± 77 % and F_{100}, +16 ± 14 %; Table 2). This resulted in a time × group interaction effect for F_{100} (Table 2; Fig. 3a) and a tendency for a time × group interaction effect for F_{50} (ANOVA, $P = 0.083$; Table 2; Fig. 3a). For F_{150} there was no time × group interaction effect (Table 2; Fig. 2a), as there were similar increases in the MST (+8 ± 10 %; paired *t* test, $P = 0.059$) and EST groups (+14 ± 8 %; Table 2).

There were time × group interaction effects for F_{50}, F_{100}, and F_{150} relative to MVF (Table 2; Fig. 3c). This was due to relative explosive force changing in opposite directions for the two groups (Fig. 3c). Specifically, EST induced an increase in F_{50} relative to MVF (+53 ± 64 %; Table 2), but no change in F_{100} or F_{150} (Table 2) relative to MVF. In contrast, after MST there was no change in F_{50}, but a decrease in F_{100} (−14 ± 10 %; Table 2) and F_{150} (−10 ± 11 %; Table 2) relative to MVF.

EMG variables

There was a main effect of time on EMG_{MVF} normalised to M_{max} (ANOVA, $P < 0.001$), and whilst there was no time × group interaction effect (Table 3; Fig. 2), post hoc analysis revealed an increase in EMG_{MVF} normalised to

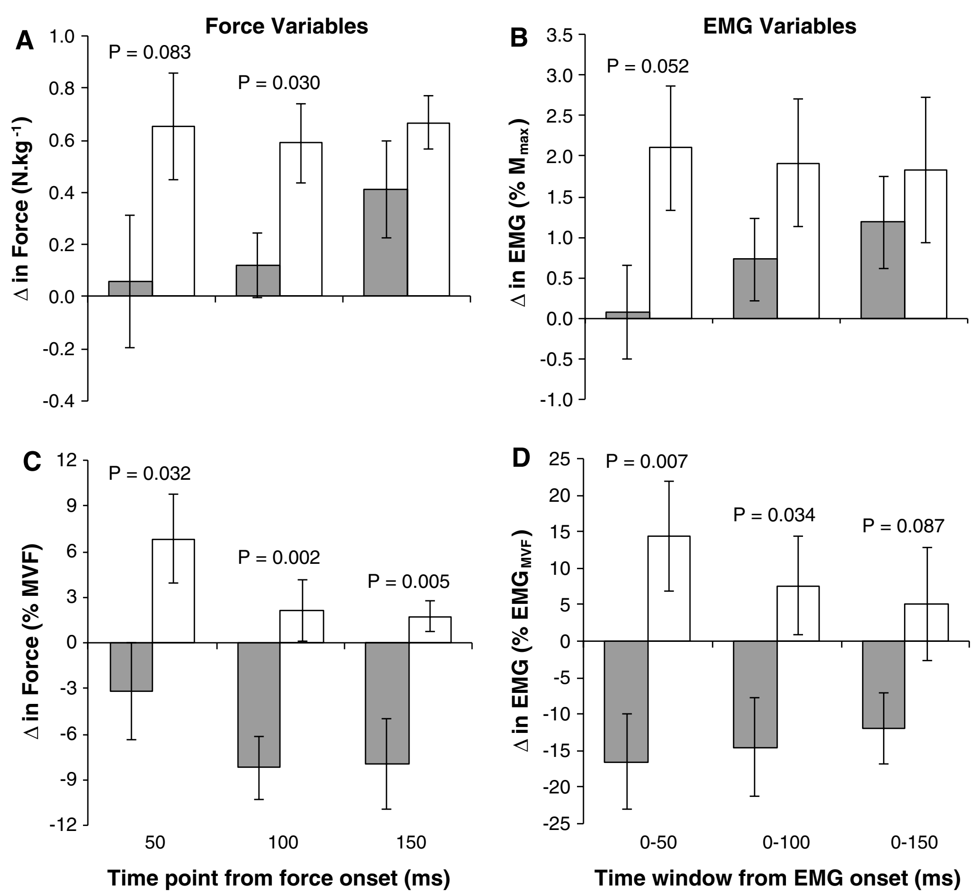


Fig. 3 Pre- to post-training changes in force production and mean quadriceps EMG during explosive isometric knee extensions in the maximal (*filled bars*) and explosive (*open bars*) strength training groups. Force is expressed relative to body mass (**a**) and maximal voluntary force (MVF; **c**), whilst EMG is expressed relative to maximal M-wave (M_{max}; **b**) and EMG at MVF (EMG_{MVF}; **d**) prior to calculating changes. The *P*-value denotes a time × group interaction effect ($P < 0.05$), or tendency for a time × group interaction effect ($0.1 > P > 0.05$) determined via a two-way ANOVA (time, pre vs. post; by group, maximal vs. explosive). Data are mean ± SEM

Table 3 EMG recorded at maximal voluntary force (EMG_{MVF}) and over different time periods from EMG onset (0–50, 0–100, and 0–150 ms) during explosive contractions performed pre- and post-training in the maximal (MST) and explosive strength training (EST) groups

	MST group		EST group		Interaction effect (*P* value)
	Pre	Post	Pre	Post	
EMG_{MVF} (%M_{max})	9.6 ± 1.9	12.3 ± 2.3*	9.6 ± 2.3	10.9 ± 2.1	0.278
$EMG_{0–50}$ (%M_{max})	6.8 ± 2.5	6.9 ± 2.2	5.0 ± 2.0	7.1 ± 2.0*	0.052
$EMG_{0–100}$ (%M_{max})	8.9 ± 2.8	9.6 ± 2.7	8.0 ± 1.9	10.0 ± 2.3*	0.228
$EMG_{0–150}$ (%M_{max})	9.5 ± 3.0	10.7 ± 2.6	8.9 ± 2.2	10.7 ± 2.3	0.565
$EMG_{0–50}$ (%EMG_{MVF})	74.7 ± 23.7	58.2 ± 17.7*	54.3 ± 20.6	68.7 ± 19.4	0.007
$EMG_{0–100}$ (%EMG_{MVF})	96.7 ± 22.7	82.1 ± 22.5	87.2 ± 19.6	94.8 ± 17.0	0.034
$EMG_{0–150}$ (%EMG_{MVF})	102.8 ± 20.6	90.8 ± 19.8*	95.8 ± 20.6	101.0 ± 15.2	0.087

All variables were normalised to maximal M-wave (M_{max}), and EMG variables during the explosive contractions are also expressed relative to EMG_{MVF}. Time (pre vs. post) × group (MST vs. EST) interaction effects are also reported. Data are mean ± SD

Within-group effects of training are denoted by * ($P < 0.05$)

M_{max} after MST (+34 ± 38 %; Table 3), but no change following EST (Table 3).

There was a main effect of time on $EMG_{0–50}$, $EMG_{0–100}$, and $EMG_{0–150}$ (ANOVAs, $0.013 \leq P \leq 0.039$). Post hoc analysis revealed an increase in $EMG_{0–50}$ (+65 ± 73 %) and $EMG_{0–100}$ (+18 ± 28 %) normalised to M_{max} (Table 3), and a tendency for an increase in $EMG_{0–150}$ (+26 ± 50 %; paired *t* tests, $P = 0.073$; Table 3), following EST. In contrast, the MST group displayed no change in $EMG_{0–50}$ or $EMG_{0–100}$ normalised to M_{max} (Table 3), but a similar increase to EST in $EMG_{0–150}$ (+26 ± 31 %: paired *t* test, $P = 0.073$; Table 3). There was a near-significant time × group interaction effect on $EMG_{0–50}$ normalised to M_{max} (ANOVA, $P = 0.052$; Table 3; Fig. 3b).

There was a time × group interaction effect on $EMG_{0–50}$ and $EMG_{0–100}$ relative to EMG_{MVF} (Table 3; Fig. 3d), and a tendency for a time × group interaction effect on $EMG_{0–150}$ relative to EMG_{MVF} (ANOVA, $P = 0.087$; Table 3; Fig. 3d), due to these variables changing in opposite directions for the two groups. Specifically, MST induced a decrease in $EMG_{0–50}$ (−10 ± 47 %; Table 3) and $EMG_{0–150}$ relative to EMG_{MVF} (−5 ± 11 %; Table 3), and a tendency for the same effect on $EMG_{0–100}$ relative to EMG_{MVF} (−10 ± 18 %; paired *t* test, $P = 0.065$; Table 3). In contrast, there was a tendency for an increase in $EMG_{0–50}$ relative to EMG_{MVF} (+43 ± 64 %; Table 3) and no change in $EMG_{0–100}$ or $EMG_{0–150}$ (Table 3) following EST.

Discussion

This investigation directly compared the functional and neural adaptations to maximal vs. explosive strength training and found distinct changes during the explosive and maximum phases of contraction that were specific to the training stimulus. MST produced greater improvements in maximal strength, and EST was more effective at improving early phase explosive strength. In addition, relative explosive strength (to MVF) showed more pronounced contrasts, with more positive changes after EST than MST. These effects appeared to be underpinned by training-specific changes in neuromuscular activation during the explosive and maximum phases of contraction. MST produced twofold greater changes in EMG_{MVF} than EST, although this difference was not significant, and EST produced greater changes in early phase neuromuscular activation during explosive contractions.

Maximal strength (MVF) increased in both groups, clearly showing that both training methods provided sufficient stimuli for functional changes at the peak of the force–time curve. Other investigations have also reported improved maximal strength following maximal strength training, explosive strength training, or a combination of both (Andersen et al. 2010; Barry et al. 2005; de Ruiter et al. 2012; Del Balso and Cafarelli 2007; Jones and Rutherford 1987; Kubo et al. 2001; Rich and Cafarelli 2000; Suetta et al. 2004). However, to the best of our knowledge this is the first study to directly compare the different training stimuli and observe greater improvements in MVF following maximal strength training than explosive strength training. Given that greater training loads are considered to be more conducive to improvements in maximal strength (Crewther et al. 2005), it is interesting that the EST group, which aimed to achieve at least 90 % MVF in the training contractions, experienced smaller maximal strength gains than the MST group, which experienced smaller training loads (75 % MVF). However, time under tension is also considered an important training stimulus for maximal strength gains (Crewther et al. 2005) and was considerably longer for MST (MST, ~4 s vs. EST, ~1 s per contraction), so may therefore be a more important determinant of maximal strength gains than the amplitude of the training load.

In contrast to the functional changes at the peak of the force–time curve, EST produced greater increases in explosive force production during the initial 100 ms of contraction than MST. Thus, explosive strength training provided an effective stimulus for improving early phase (first 0–100 ms) explosive force production, whilst maximal strength training did not. Previous investigations have consistently reported improvements in explosive force when training this component of strength via short (≤1-s), rapid contractions (Barry et al. 2005; de Ruiter et al. 2012; Gruber et al. 2007; Van Cutsem et al. 1998). In comparison, training involving more prolonged contractions (>2 s) and thus more closely associated with the maximal strength training protocol in the current study have reported inconsistent effects on explosive strength (Andersen et al. 2010; Del Balso and Cafarelli 2007; Kubo et al. 2001; Rich and Cafarelli 2000; Suetta et al. 2004). These studies employed a variety of different training protocols (e.g. intensity, volume, and duration) and may have used different instructions during the training contractions (e.g. "push fast and hard" vs. "push hard"), which would be expected to alter the training stimulus and result in differential changes in explosive strength. In the current study the EST group were instructed to push as "fast and hard" as possible, whilst the MST group received no instruction to produce force rapidly during the training contractions. Therefore, our results provide direct evidence that training contractions must be performed 'explosively' with an emphasis on high rate of force development to elicit short-term improvements in explosive strength.

The increase in EMG_{MVF} was only significant following MST, which induced a twofold greater change than after EST. On the other hand, there was no time × group interaction effect which would suggest the groups had similar changes in EMG_{MVF}. The high between-day variability in EMG even after normalisation to M_{max} (Buckthorpe et al. 2012), may have reduced the chances of observing a significant group × time interaction effect and/or a significant change in EMG_{MVF} after EST, and so these results should be interpreted with caution. Nevertheless, it seems likely that there were improvements in neuromuscular activation at MVF in both groups which contributed to their increases in MVF, but that these neural adaptations were only large enough to render a significant change in EMG_{MVF} in the MST group. Thus, it appears the MST provides a greater stimulus for changes in neural drive during the maximum (plateau) phase of contraction, which would explain the greater increases in MVF after MST compared to EST. During the early phase of explosive contraction (0–50 ms), neuromuscular activation increased more after EST than MST, and thus appears to explain the greater improvements in explosive force after EST. Therefore, in overview this study indicates that the training-specific functional changes in the response to explosive and maximal strength training are underpinned by changes in neuromuscular activation specific to the training performed. Whilst past research has linked short-term (within 4 weeks) improvements in maximal and explosive strength to increased neuromuscular activation during the maximum (Del Balso and Cafarelli 2007; Tillin et al. 2011) and explosive (Barry et al. 2005; de Ruiter et al. 2012; Tillin et al. 2012b) phases of contraction, respectively, to the best of our knowledge, this is the first study to directly show that distinct training stimuli are required to elicit these adaptations.

The distinct functional and neural adaptations to training for maximal vs. explosive strength are emphasised when considering changes during the rising force–time curve relative to those at the peak of the force–time curve (Fig. 3c, d). Specifically, the EST group showed increases in relative explosive force and EMG, due to improvements in neuromuscular activation during the explosive phase of contraction that did not transfer to the maximum phase of contraction. In contrast, the MST group showed a decrease in relative explosive force and EMG, due to improvements in neuromuscular activation during the maximum phase of contraction that did not transfer to the explosive phase of contraction. These disproportionate changes during the explosive and maximum phases of contraction for both types of training provide evidence that the capacity to voluntarily activate the muscles and produce force during the rising force–time curve can be trained independently of the capacity to voluntarily activate the muscles and produce force at the peak of the force–time curve, and vice versa. This is a particularly interesting finding considering there is good evidence to suggest that a similar mechanism, specifically increased motor unit firing frequency, is likely to be the dominant adaptive response underpinning improvements in neural drive during both the explosive (Van Cutsem et al. 1998) and maximum (Kamen and Knight 2004, 2008) phases of contraction (Duchateau et al. 2006).

Previous investigations have reported evidence of morphological adaptations (e.g. increased muscle size, shifts in fibre type, and increased electrically evoked force response) within just 4 weeks of strength training (Jurimae et al. 1996; Seynnes et al. 2007; Staron et al. 1994; Tillin et al. 2012b), so we cannot rule out the possibility that morphological adaptations contributed to the training × group interaction effects observed in the current study. However, given there was a close association between neural and functional adaptations, we believe the contribution of any morphological adaptations to the observed interaction effects were minimal.

In conclusion, short-term training for maximal or explosive strength elicited distinct functional and neural adaptations which were specific to the training stimulus. There were greater improvements in maximal strength in the MST group, which appeared due to their twofold greater improvements in neuromuscular activation at MVF, whilst early phase explosive strength increased in the EST group only, due to greater improvements in early phase neuromuscular activation following EST. These results demonstrate the independent adaptability of maximal and explosive strength in response to training.

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References

Ada L, Canning C, Dwyer T (2000) Effect of muscle length on strength and dexterity after stroke. Clin Rehabil 14:55–61

Allison GT (2003) Trunk muscle onset detection technique for EMG signals with ECG artefact. J Electromyogr Kinesiol 13:209–216

Andersen LL, Aagaard P (2006) Influence of maximal muscle strength and intrinsic muscle contractile properties on contractile rate of force development. Eur J Appl Physiol 96:46–52. doi:10.1007/s00421-005-0070-z

Andersen LL, Andersen JL, Zebis MK, Aagaard P (2010) Early and late rate of force development: differential adaptive responses to resistance training? Scand J Med Sci Sports 20:162–169. doi:10.1111/j.1600-0838.2009.00933.x

Barry BK, Warman GE, Carson RG (2005) Age-related differences in rapid muscle activation after rate of force development training of the elbow flexors. Exp Brain Res 162:122–132. doi:10.1007/s00221-004-2127-3

Bojsen-Moller J, Magnusson SP, Rasmussen LR, Kjaer M, Aagaard P (2005) Muscle performance during maximal isometric and dynamic contractions is influenced by the stiffness of the tendinous structures. J Appl Physiol 99:986–994. doi:10.1152/japplphysiol.01305.2004

Buckthorpe MW, Hannah R, Pain MTG, Folland JP (2012) Reliability of neuromuscular measurements during explosive isometric contractions, with special reference to electromyography normalization techniques. Muscle Nerve 46:566–576. doi:10.1002/mus.23322

Crewther B, Cronin J, Keogh J (2005) Possible stimuli for strength and power adaptation: acute mechanical responses. Sports Med 35:967–989

de Ruiter CJ, Van Leeuwen D, Heijblom A, Bobbert MF, de Haan A (2006) Fast unilateral isometric knee extension torque development and bilateral jump height. Med Sci Sports Exerc 38:1843–1852. doi:10.1249/01.mss.0000227644.14102.50

de Ruiter CJ, Hutter V, Icke C et al (2012) The effects of imagery training on fast isometric knee extensor torque development. J Sports Sci 30:166–174. doi:10.1080/02640414.2011.627369

Del Balso C, Cafarelli E (2007) Adaptations in the activation of human skeletal muscle induced by short-term isometric resistance training. J Appl Physiol 103:402–411. doi:10.1152/japplphysiol.00477.2006

Domire ZJ, Boros RL, Hashemi J (2011) An examination of possible quadriceps force at the time of anterior cruciate ligament injury during landing: a simulation study. J Biomech 44:1630–1632. doi:10.1016/j.jbiomech.2011.03.001

Duchateau J, Semmler JG, Enoka RM (2006) Training adaptations in the behavior of human motor units. J Appl Physiol 101:1766–1775. doi:10.1152/japplphysiol.00543.2006

Faulkner JA, Larkin LM, Claflin DR, Brooks SV (2007) Age-related changes in the structure and function of skeletal muscles. Clin Exp Pharmacol Physiol 34:1091–1096. doi:10.1111/j.1440-1681.2007.04752.x

Folland JP, Williams AG (2007) The adaptations to strength training: morphological and neurological contributions to increased strength. Sports Med 37:145–168

Garcia-Pallares J, Lopez-Gullon JM, Muriel X, Diaz A, Izquierdo M (2011) Physical fitness factors to predict male Olympic wrestling performance. Eur J Appl Physiol 111:1747–1758. doi:10.1007/s00421-010-1809-8

Gruber M, Gruber SB, Taube W, Schubert M, Beck SC, Gollhofer A (2007) Differential effects of ballistic versus sensorimotor training on rate of force development and neural activation in humans. J Strength Cond Res 21:274–282. doi:10.1519/R-20085.1

Hakkinen K, Keskinen KL (1989) Muscle cross-sectional area and voluntary force production characteristics in elite strength- and endurance-trained athletes and sprinters. Eur J Appl Physiol Occup Physiol 59:215–220

Jones DA, Parker DF (1989) Development of a portable strain gauge to measure human muscle isometric strength. J Physiol 145:11P

Jones DA, Rutherford OM (1987) Human muscle strength training: the effects of three different regimens and the nature of the resultant changes. J Physiol 391:1–11

Jurimae J, Abernethy PJ, Blake K, McEniery MT (1996) Changes in the myosin heavy chain isoform profile of the triceps brachii muscle following 12 weeks of resistance training. Eur J Appl Physiol Occup Physiol 74:287–292

Kamen G, Knight CA (2004) Training-related adaptations in motor unit discharge rate in young and older adults. J Gerontol A Biol Sci Med Sci 59:1334–1338

Knight CA, Kamen G (2008) Relationships between voluntary activation and motor unit firing rate during maximal voluntary contractions in young and older adults. Eur J Appl Physiol 103:625–630. doi:10.1007/s00421-008-0757-z

Kubo K, Kanehisa H, Ito M, Fukunaga T (2001) Effects of isometric training on the elasticity of human tendon structures in vivo. J Appl Physiol 91:26–32

Moretti DV, Babiloni F, Carducci F et al (2003) Computerized processing of EEG–EOG–EMG artifacts for multi-centric studies in EEG oscillations and event-related potentials. Int J Psychophysiol 47:199–216

Pain MTG, Hibbs A (2007) Sprint starts and the minimum auditory reaction time. J Sports Sci 25:79–86. doi:10.1080/02640410600718004

Pijnappels M, van der Burg PJ, Reeves ND, van Dieen JH (2008) Identification of elderly fallers by muscle strength measures. Eur J Appl Physiol 102:585–592. doi:10.1007/s00421-007-0613-6

Pulkovski N, Schenk P, Maffiuletti NA, Mannion AF (2008) Tissue Doppler imaging for detecting onset of muscle activity. Muscle Nerve 37:638–649. doi:10.1002/mus.20996

Quarrie KL, Wilson BD (2000) Force production in the rugby union scrum. J Sports Sci 18:237–246. doi:10.1080/026404100364974

Rich C, Cafarelli E (2000) Submaximal motor unit firing rates after 8 wk of isometric resistance training. Med Sci Sports Exerc 32:190–196

Sahaly R, Vandewalle H, Driss T, Monod H (2001) Maximal voluntary force and rate of force development in humans—importance of instruction. Eur J Appl Physiol 85:345–350

Sale DG (2003) Neural adaptations to strength training. In: Komi PV (ed) Strength and power in sport, 2nd edn. Blackwell Science Ltd, Oxford

Schantz P, Randall-Fox E, Hutchison W, Tyden A, Astrand PO (1983) Muscle fibre type distribution, muscle cross-sectional area and maximal voluntary strength in humans. Acta Physiol Scand 117:219–226

Seynnes OR, de Boer M, Narici MV (2007) Early skeletal muscle hypertrophy and architectural changes in response to high-intensity resistance training. J Appl Physiol 102:368–373. doi:10.1152/japplphysiol.00789.2006

Staron RS, Karapondo DL, Kraemer WJ et al (1994) Skeletal muscle adaptations during early phase of heavy-resistance training in men and women. J Appl Physiol 76:1247–1255

Suetta C, Aagaard P, Rosted A et al (2004) Training-induced changes in muscle CSA, muscle strength, EMG, and rate of force development in elderly subjects after long-term unilateral disuse. J Appl Physiol 97:1954–1961. doi:10.1152/japplphysiol.01307.2003

Thorstensson A, Karlsson J, Viitasalo JH, Luhtanen P, Komi PV (1976) Effect of strength training on EMG of human skeletal muscle. Acta Physiol Scand 98:232–236

Tillin NA, Jimenez-Reyes P, Pain MTG, Folland JP (2010) Neuromuscular performance of explosive power athletes versus untrained individuals. Med Sci Sports Exerc 42:781–790. doi:10.1249/MSS.0b013e3181be9c7eER

Tillin NA, Pain MTG, Folland JP (2011) Short-term unilateral resistance training affects the agonist–antagonist but not the force–agonist activation relationship. Muscle Nerve 43:375–384. doi:10.1002/mus.21885

Tillin NA, Pain MTG, Folland JP (2012a) Contraction type influences the human ability to use the available torque capacity of skeletal muscle during explosive efforts. Proc Biol Sci 279:2106–2115. doi:10.1098/rspb.2011.2109

Tillin NA, Pain MTG, Folland JP (2012b) Short-term training for explosive strength causes neural and mechanical adaptations. Exp Physiol 97:630–641. doi:10.1113/expphysiol.2011.063040

Tillin NA, Pain MTG, Folland J (2013a) Explosive force production during isometric squats correlates with athletic performance in rugby union players. J Sports Sci 31:66–76. doi:10.1080/02640414.2012.720704

Tillin NA, Pain MTG, Folland JP (2013b) Identification of contraction onset during explosive contractions. Response to Thompson et al. "Consistency of rapid muscle force characteristics: influence of muscle contraction onset detection methodology" (J Electromyogr Kinesiol 2012;22:893–900). J Electromyogr Kinesiol 23:991–994. doi:10.1016/j.jelekin.2013.04.015

Van Cutsem M, Duchateau J, Hainaut K (1998) Changes in single motor unit behaviour contribute to the increase in contraction speed after dynamic training in humans. J Physiol 513:295–305